

Assignments for Chapter 1

- 1.20
- 1.21 (c) (f)
- 1.24 (a)
- 1.26
- 1.27 (a) (f)
- 1.41

Tutorial Questions (Week 3)

- **Basic Problems with Answers 1.15, 1.18**
- **Basic Problems 1.29, 1.31**
- **Advanced Problems 1.33, 1.42**

- **1教107**
- **周1、2、3、4晚上9:00-10:00**

Signals and Systems from Our Point of View

- **Signal** can be defined as an **observable** change in a **quantifiable** entity.
- **Signal** is able to carry information on the nature of a physical phenomenon.
- **Signals** can be represented by functions, but they are *not equivalent to functions* (signals are physical).
- **Systems** process input signals to produce output signals.
- **Systems** can be viewed as **mapping** between input and output signals.
- The course is about using **mathematical** techniques to analyze and synthesize systems which process signals.

[*] P. Chakravorty, "What Is a Signal? [Lecture Notes]," *IEEE Signal Processing Magazine*, vol. 35, no. 5, pp. 175-177, Sept. 2018

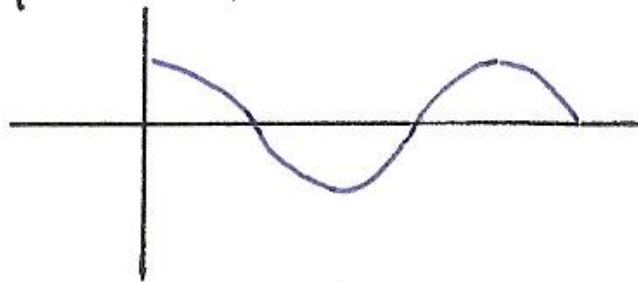
Periodic Complex Exponential/Sinusoidal Signals

Two expressions for complex numbers

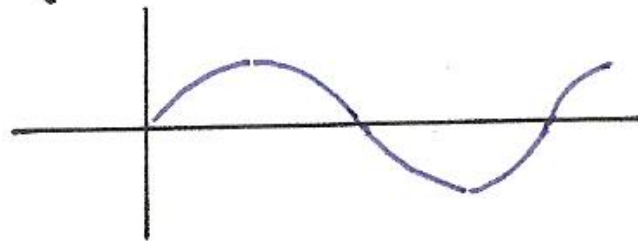
$$z = Ae^{j\theta} = A\cos\theta + jA\sin\theta$$

$$x(t) = e^{j\omega_0 t} = \cos(\omega_0 t) + j\sin(\omega_0 t)$$

$$\operatorname{Re}\{e^{j\omega_0 t}\} = \cos\omega_0 t$$



$$\operatorname{Im}\{e^{j\omega_0 t}\} = \sin\omega_0 t$$



-Fundamental frequency: ω_0

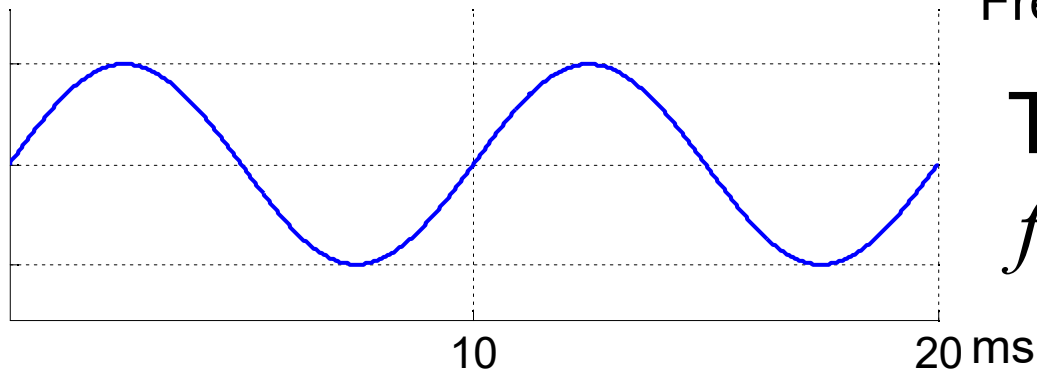
-Fundamental period: $T_0 = \frac{2\pi}{\omega_0}$

-In CT, $e^{j\omega_0 t}$ **always** periodic

-Distinct signals for distinct values of ω_0 .

-Rapid variation with large ω_0

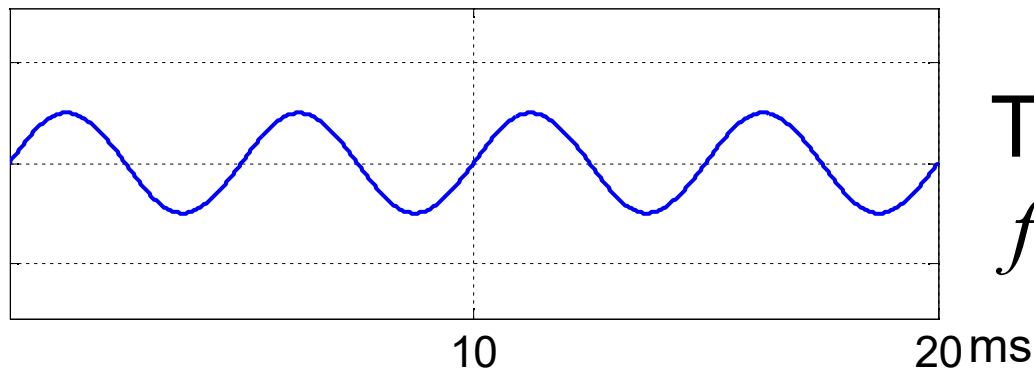
Harmonically Related Signal Sets



$$\text{Frequency} \leftarrow f = \frac{\omega}{2\pi} \rightarrow \text{Angular Frequency (rad/s)}$$

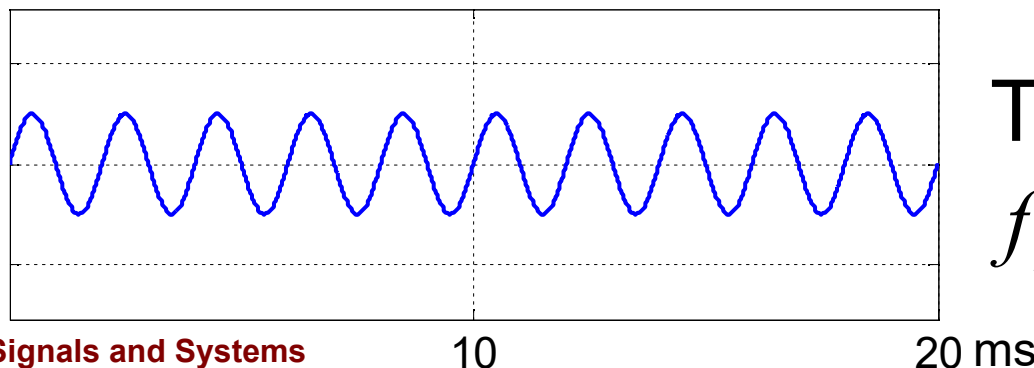
$$T_0 = 10 \text{ ms} \quad 1 \text{ Hz} = 2\pi \text{ rad/s}$$

$$f_0 = 100 \text{ Hz}$$



$$T_1 = 5 \text{ ms}$$

$$f_1 = 200 \text{ Hz}$$



$$T_2 = 2 \text{ ms}$$

$$f_2 = 500 \text{ Hz}$$

Harmonically Related Signal Sets (cont.)

- A set of periodic exponentials which have a common period T_0 .

$$\{\phi_k(t) = e^{jk\omega_0 t}, k = 0, \pm 1, \pm 2, \dots\}$$

fundamental frequency $|k\omega_0|$

must be integer multiple

fundamental period $T_k = \frac{2\pi}{|k\omega_0|} = \frac{T_0}{|k|}, \quad T_0 = \frac{2\pi}{\omega_0}$

- The k th harmonic $\phi_k(t)$ is periodic with period T_0 , as it goes through $|k|$ of its fundamental period T_k in duration of length T_0 .

Periodicity Properties of DT Complex Exponentials

Important Differences Between Continuous-time and Discrete-time Exponential/Sinusoidal Signals

- For discrete-time, signals with frequencies ω_0 and $\omega_0 + m \cdot 2\pi$ are identical. This is Not true for continuous-time.

Proof:

$$e^{j(\omega_0 + m \cdot 2\pi)n} = e^{j\omega_0 n} \cdot e^{jm \cdot 2\pi n} = e^{j\omega_0 n}$$

$$\forall x \neq 0, e^{j(\omega_0 + x)t} = e^{j\omega_0 t} e^{jxt} \neq e^{j\omega_0 t}$$

Periodicity Properties of DT Complex Exponentials (cont.)

Understanding:

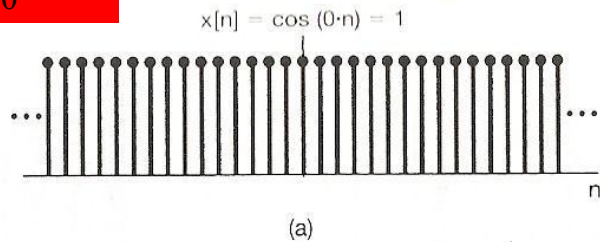
- We need only consider a frequency interval of length 2π , and in most cases, we use the interval: $0 \leq \omega_0 < 2\pi$, or $-\pi \leq \omega_0 < \pi$
- $e^{j\omega_0 n}$ does **not** have a continuously increasing rate of oscillation as ω_0 is increased in magnitude.
 - low-frequency (slowly varying): ω_0 near $0, 2\pi, \dots$, or $2k \cdot \pi$
 - high-frequency (rapidly varying): ω_0 near $\pm \pi, \dots$, or $(2k+1) \cdot \pi$

$$e^{j(2k+1)\pi n} = e^{j\pi n} = (e^{j\pi})^n = (-1)^n$$

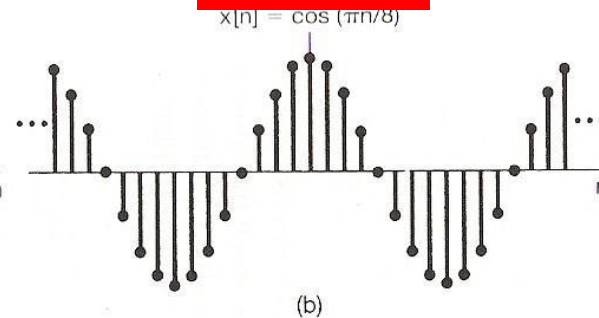
$$e^{j2k\pi n} = (e^{j2\pi})^{kn} = (1)^{kn} = 1$$

Review

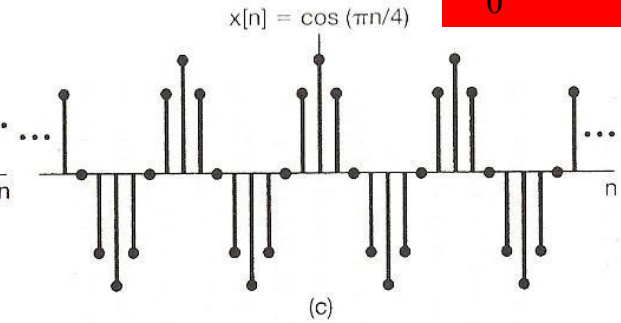
$$\omega_0 = 0$$



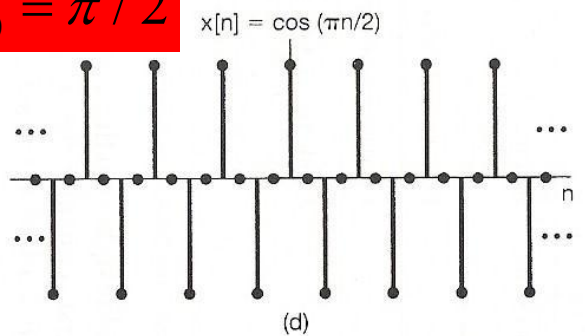
$$\omega_0 = \pi / 8$$



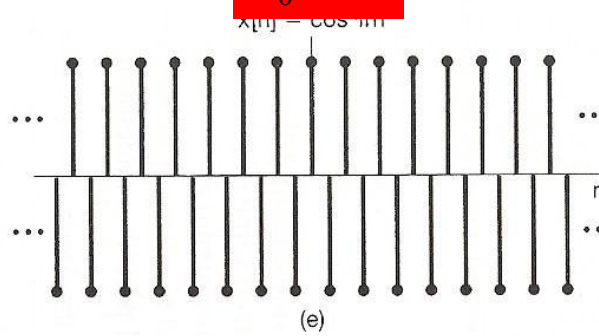
$$\omega_0 = \pi / 4$$



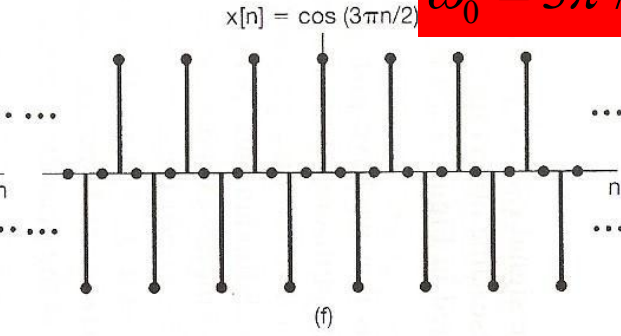
$$\omega_0 = \pi / 2$$



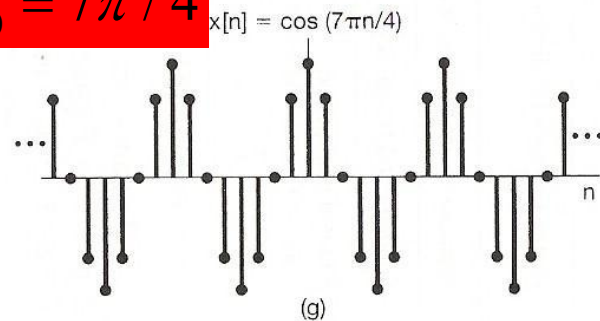
$$\omega_0 = \pi$$



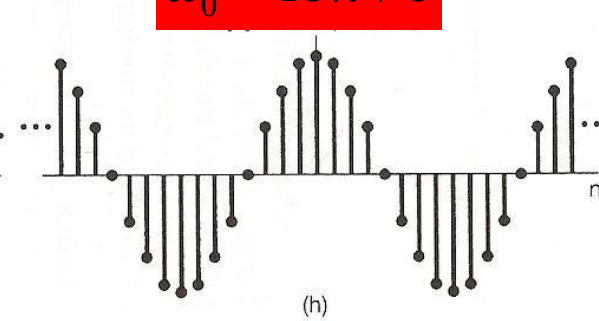
$$\omega_0 = 3\pi / 2$$



$$\omega_0 = 7\pi / 4$$



$$\omega_0 = 15\pi / 8$$



$$\omega_0 = 2\pi$$

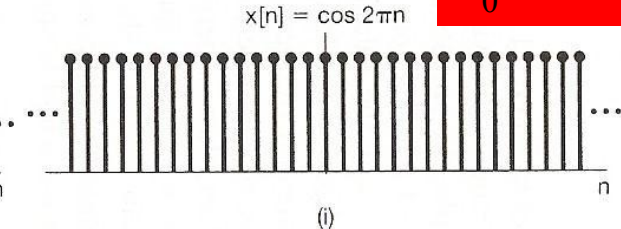


Figure 1.27 Discrete-time sinusoidal sequences for several different frequencies.

Periodicity Properties of DT Complex Exponentials (cont.)

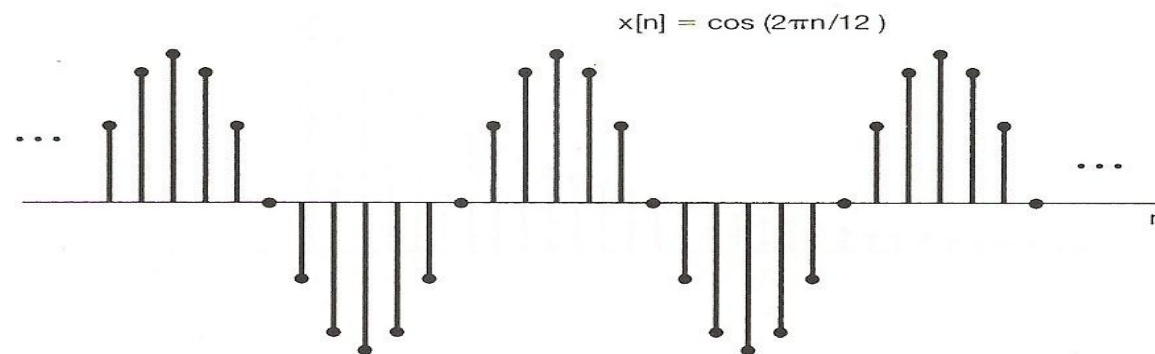
Important Differences Between Continuous-time and Discrete-time Exponential/Sinusoidal Signals

- For discrete-time, ω_0 is usually defined only for $[-\pi, \pi]$ or $[0, 2\pi]$. For continuous-time, ω_0 is defined for $(-\infty, \infty)$
- For discrete-time, the signal is periodic only when $\omega_0 N = 2\pi m$

Why?

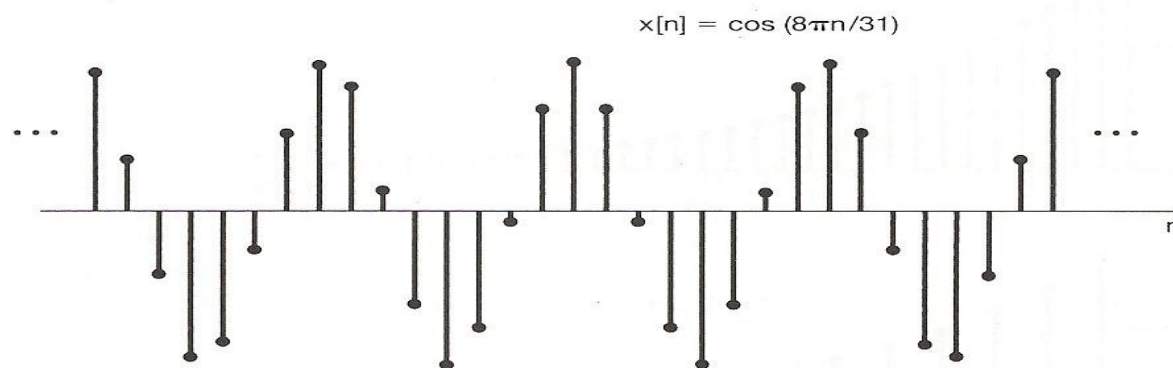
$$e^{j\omega_0 n} = e^{j\omega_0 (n+N)} \rightarrow e^{j\omega_0 N} = 1 \rightarrow \omega_0 N = 2\pi m$$

$$\frac{\omega_0}{2\pi} = \frac{m}{N} \quad \text{should be a rational number!}$$



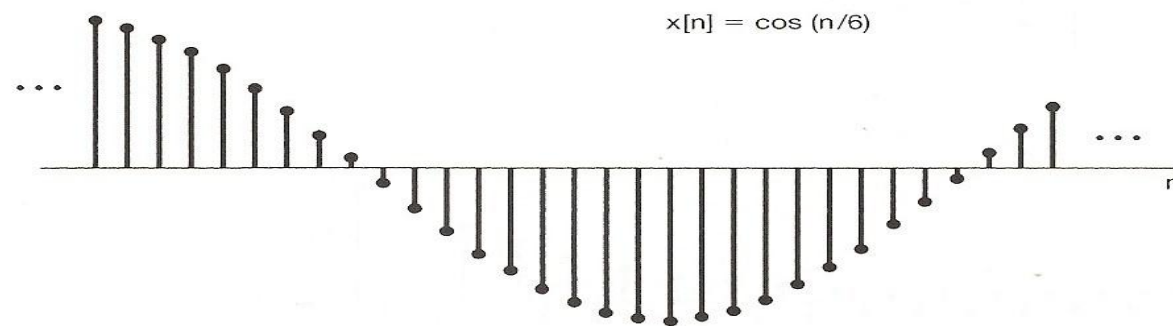
(a)

$$\frac{\omega_0}{2\pi} = \frac{2\pi/12}{2\pi} = \frac{1}{12}$$



(b)

$$\frac{\omega_0}{2\pi} = \frac{8\pi/31}{2\pi} = \frac{4}{31}$$



(c)

$$\frac{\omega_0}{2\pi} = \frac{1/6}{2\pi} = \frac{1}{12\pi}$$

Figure 1.25 Discrete-time sinusoidal signals.

DT Harmonically Related Set

Harmonically related discrete-time signal sets

$$\{\phi_k[n] = e^{jk(\frac{2\pi}{N})n}, \quad k = 0, \pm 1, \pm 2, \dots\}$$

all with common period N

There are only N elements in the above set.

Proof:
$$\phi_{k+N}[n] = e^{j(k+N)(\frac{2\pi}{N})n} = e^{jk(\frac{2\pi}{N})n} \cdot e^{j2\pi n} = e^{jk(\frac{2\pi}{N})n} = \phi_k[n]$$

This is different from continuous case. Only N distinct signals in this set.

Periodicity Properties of DT Complex Exponentials (cont.)

- How to determine the fundamental frequency and fundamental period of a periodic signal $e^{j\omega_0 n}$?

- ◆ This is different from CT periodic signal $e^{j\omega_0 t}$

- ◆ We have $\omega_0 N = 2\pi m$

Fundamental frequency: $\frac{2\pi}{N} = \frac{\omega_0}{m}$

Fundamental period: $N = m\left(\frac{2\pi}{\omega_0}\right)$

See comparison between CT and DT signals in Table 1.1

Example

- What is the fundamental period of $e^{j\frac{6}{5}\pi n}$?

$$\left\{ \frac{2\pi}{\omega_0} m \mid \forall \text{ integer } m \right\} = \left\{ \frac{5}{3} m \mid \forall \text{ integer } m \right\}$$

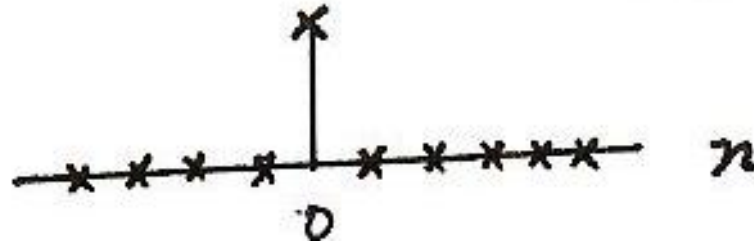
$$= \{ \dots, 0, \frac{5}{3}, \frac{10}{3}, 5, \frac{20}{3}, \dots \}$$

Hence, the fundamental period is 5 and fundamental frequency is $\frac{2\pi}{5}$.

Unit Impulse (or Unit Sample) Function

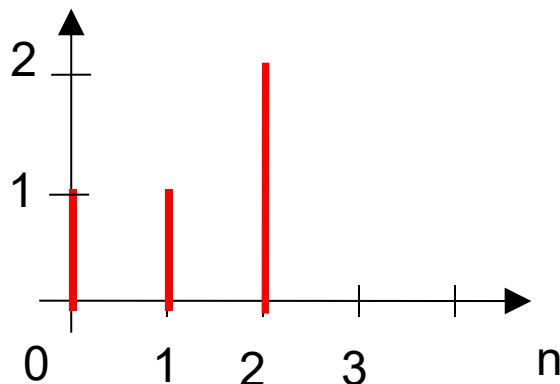
Discrete-time

$$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$



- As a basic building function, we can use unit impulse function to represent other different signals.

e.g.1



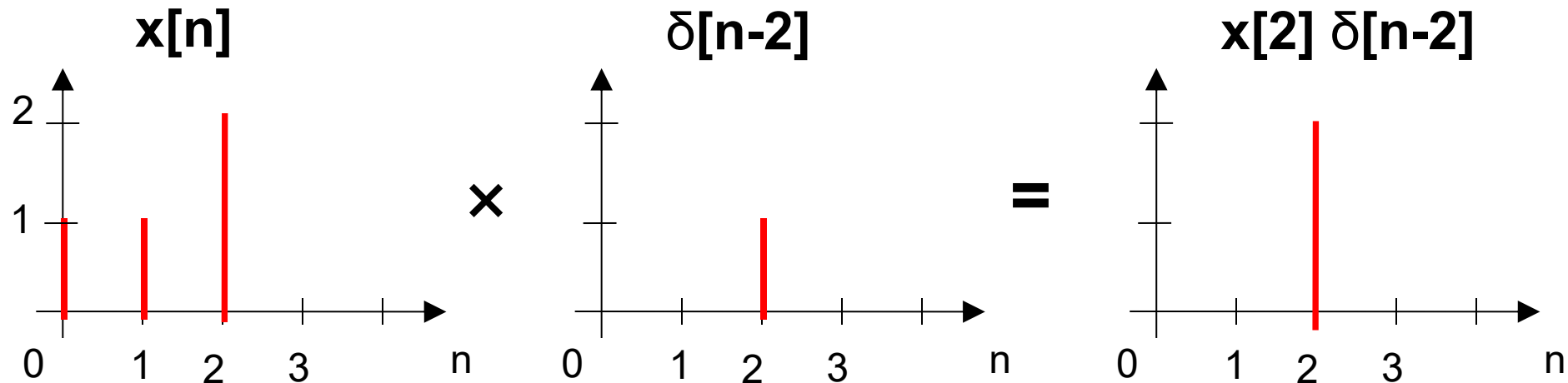
$$= \delta[n] + \delta[n - 1] + 2\delta[n - 2]$$

Unit Impulse Function (cont.)

- Sampling property

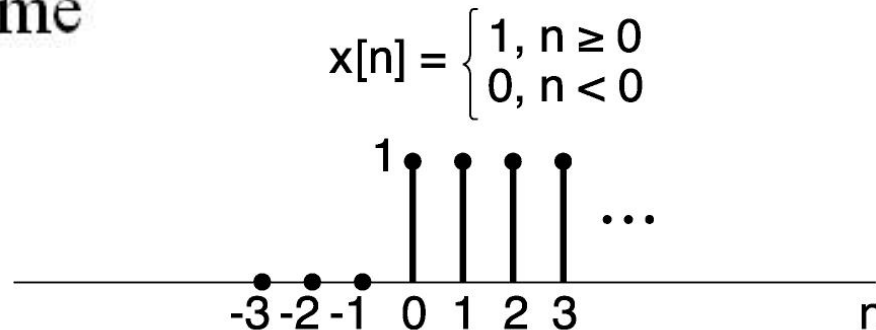
$$x[n] \delta[n] = x[0] \delta[n]$$

$$x[n] \delta[n-n_0] = x[n_0] \delta[n-n_0]$$



Unit Step Function

Discrete-time



Relation between unit impulse and unit step functions

– First difference

$$\delta[n] = u[n] - u[n-1]$$

– Running Sum

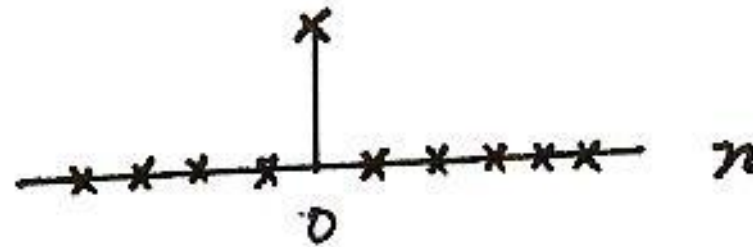
$$u[n] = \sum_{m=-\infty}^n \delta[m] \quad \left\{ \begin{array}{l} =0, \quad n < 0 \\ =1, \quad n \geq 0 \end{array} \right.$$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$

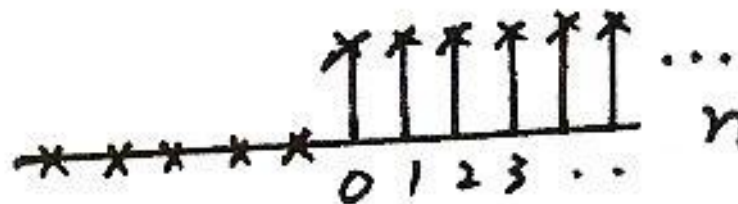
Unit Step Function: First Difference

Discrete-time

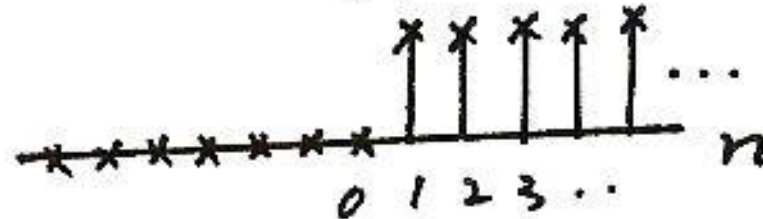
$\delta[n]$



$u[n]$



$u[n-1]$



$$\delta[n] = u[n] - u[n-1]$$

Unit Step Function: Running Sum

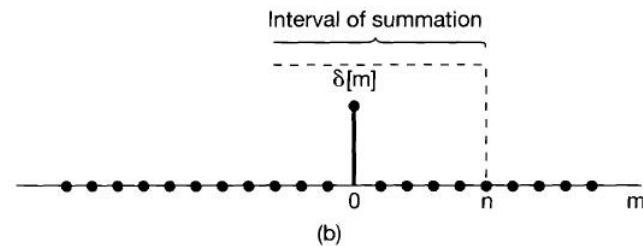
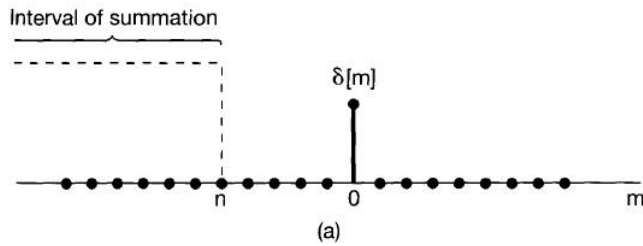


Figure 1.30 Running sum of eq. (1.66): (a) $n < 0$; (b) $n > 0$.

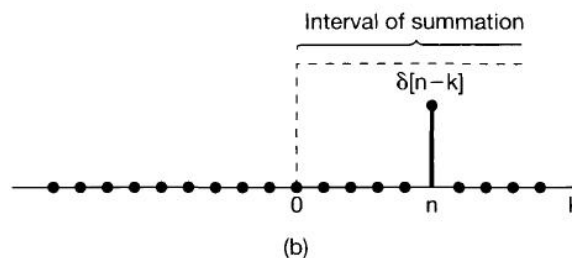
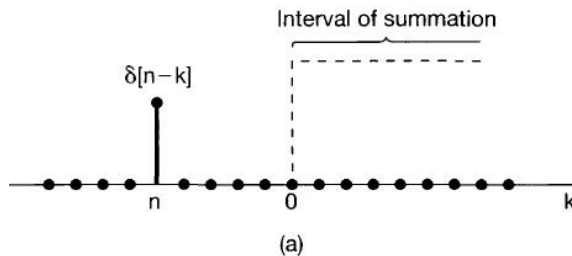


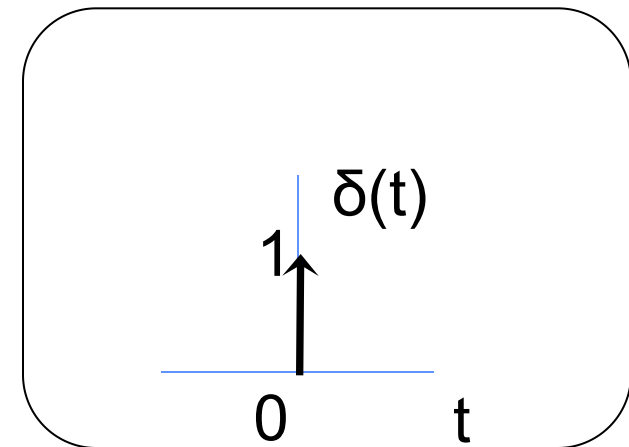
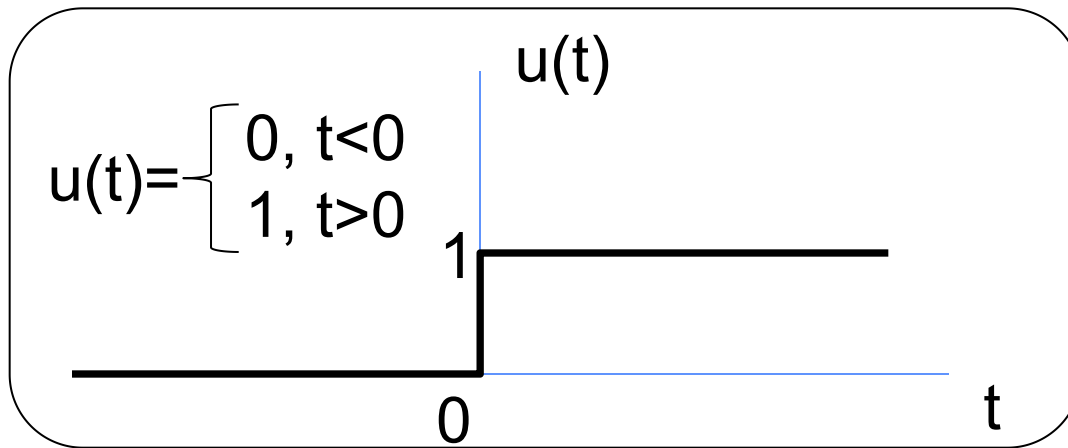
Figure 1.31 Relationship given in eq. (1.67): (a) $n < 0$; (b) $n > 0$.

$$u[n] = \sum_{m=-\infty}^n \delta[m].$$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k].$$

Unit Impulse and Unit Step Functions

Continuous-time



Relation between unit impulse and unit step functions

– First Derivative

$$\delta(t) = \frac{du(t)}{dt}$$

– Running Integral

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

Unit Impulse and Unit Step Functions (Cont.)

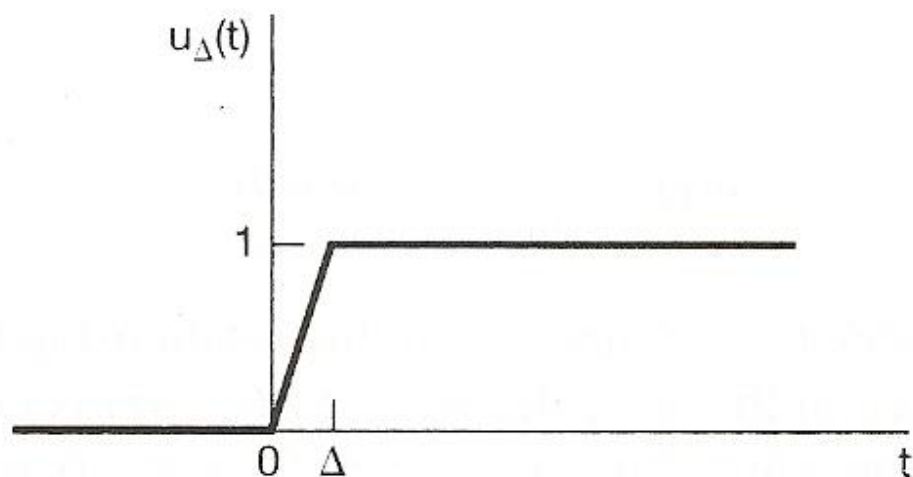


Figure 1.33 Continuous approximation to the unit step, $u_{\Delta}(t)$.

$$u(t) = \lim_{\Delta \rightarrow 0} u_{\Delta}(t)$$

$$\delta_{\Delta}(t) = \frac{du_{\Delta}(t)}{dt}$$

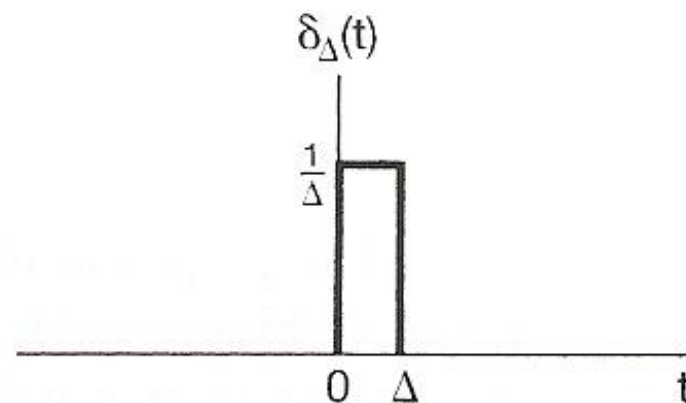


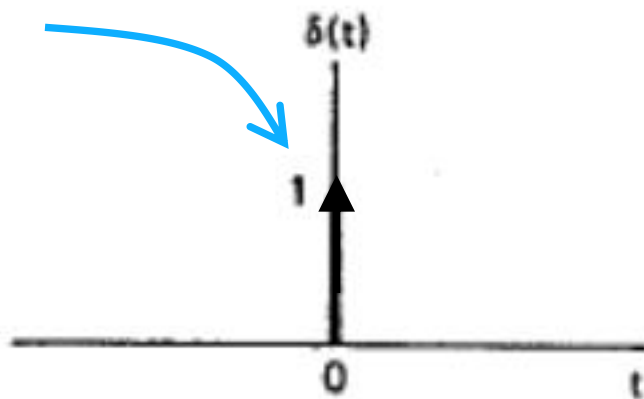
Figure 1.34 Derivative of $u_{\Delta}(t)$.

$$\delta(t) = \lim_{\Delta \rightarrow 0} \delta_{\Delta}(t)$$

More on CT unit impulse function:

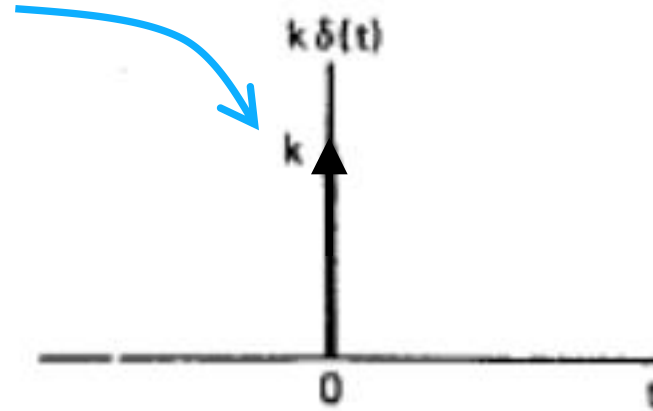
- $\delta(t)$ has in effect no duration, but unit area.

area = 1



CT unit impulse

area = k



Scaled impulse

- Or the integration of CT unit impulse function is unit.

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

More on CT unit impulse function:

Sampling Property:

$$x(t)\delta(t) = x(0)\delta(t).$$

$$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0).$$

Running Integral:

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau = \int_{\infty}^0 \delta(t - \sigma)(-d\sigma),$$

$$u(t) = \int_0^{\infty} \delta(t - \sigma) d\sigma.$$

$$\sigma = t - \tau$$

More on CT unit impulse function:

Running Integral:

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau = \int_{\infty}^0 \delta(t - \sigma)(-d\sigma),$$

$$u(t) = \int_0^{\infty} \delta(t - \sigma) d\sigma.$$

$$\sigma = t - \tau$$

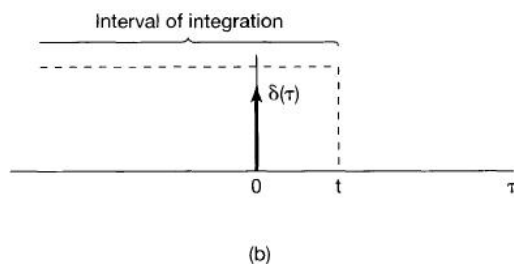
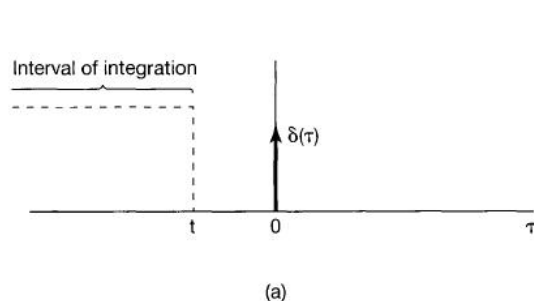


Figure 1.37 Running integral given in eq. (1.71):
(a) $t < 0$; (b) $t > 0$.

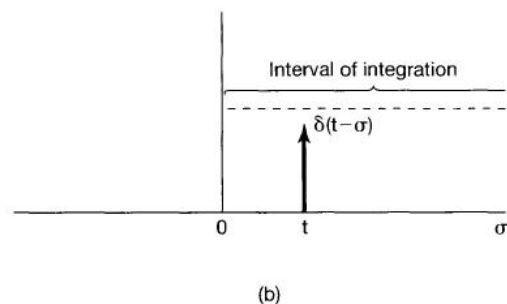
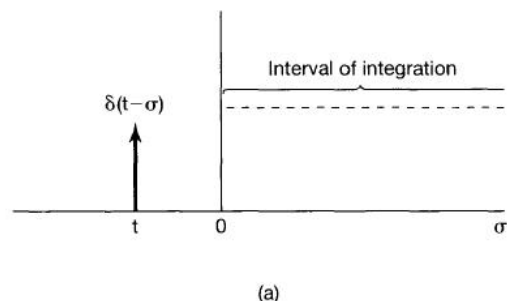
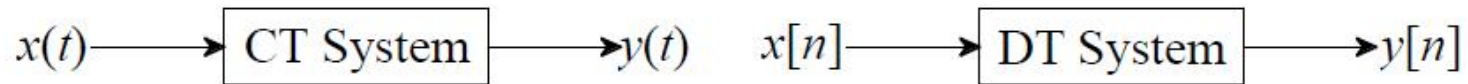


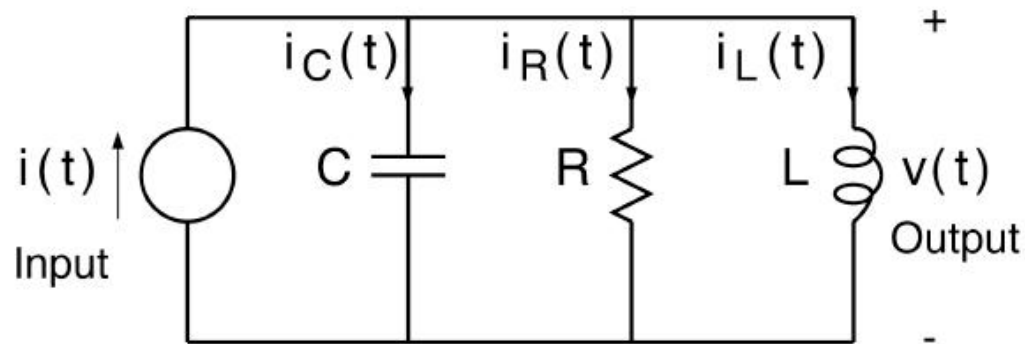
Figure 1.38 Relationship given in eq. (1.75):
(a) $t < 0$; (b) $t > 0$.

System Examples

- Systems are described from input/output perspective, that is, input to the system x causes the output y



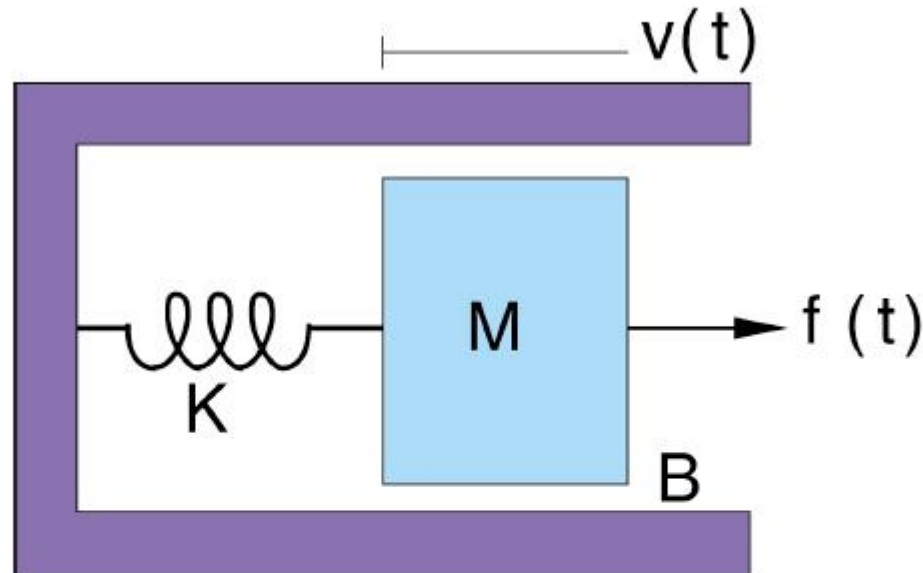
Ex. #1 RLC circuit — an electrical system



$$i(t) = \underbrace{C \frac{dv(t)}{dt}}_{\text{capacitance}} + \underbrace{\frac{v(t)}{R}}_{\text{resistance}} + \underbrace{\frac{1}{L} \int_{-\infty}^t v(\tau) d\tau}_{\text{inductance}}.$$

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Ex. #2 A shock absorber – a mechanical system

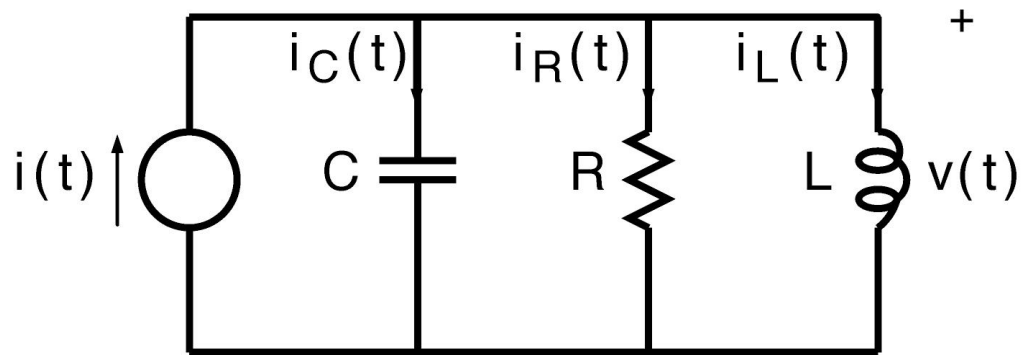
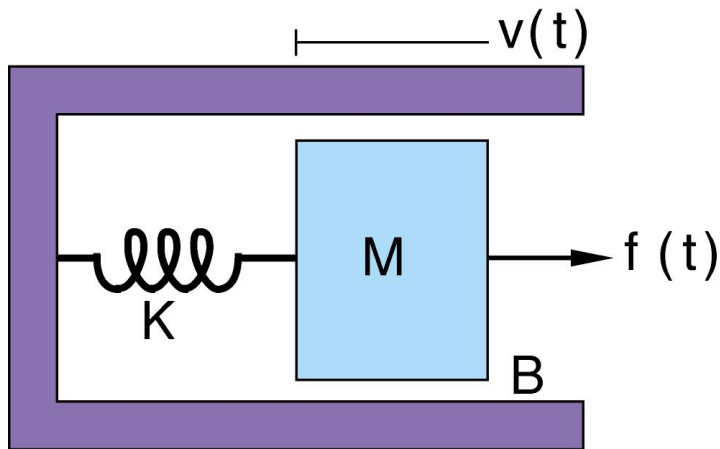


Force Balance: 慣性力 摩擦力 彈力

$$f(t) = \underbrace{M \frac{dv(t)}{dt}}_{\text{inertial force}} + \underbrace{Bv(t)}_{\text{friction}} + \underbrace{K \int_{-\infty}^t v(\tau) d\tau}_{\text{spring force}} .$$

This equation looks quite familiar, we just saw it earlier.

- **Observation: different systems could be described by the same input/output relations**

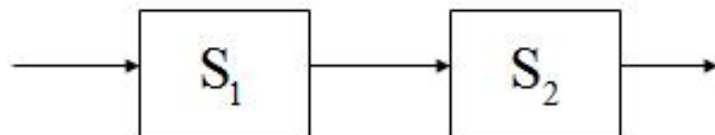


Observations

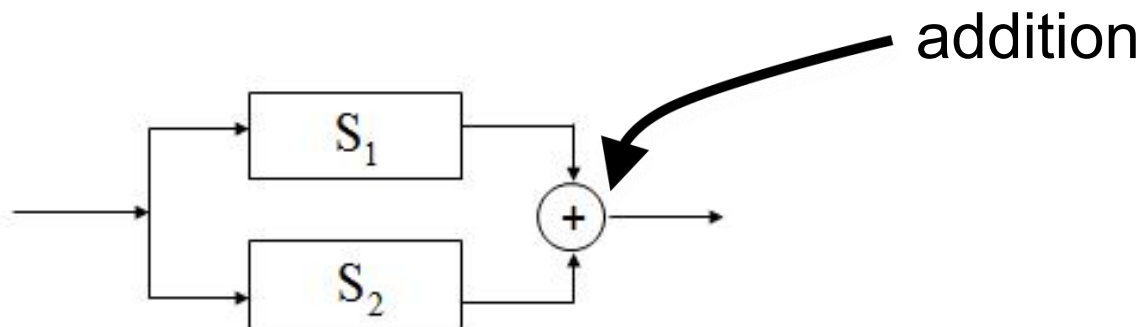
- A very rich class of systems are described by **differential/difference equations**.
- Such an equation, by itself, does not completely describe the input-output behaviour of a system: we need **auxiliary conditions** (initial conditions, boundary conditions).
- Very different physical systems may have very similar or same **mathematical descriptions**.

Interconnection of Systems

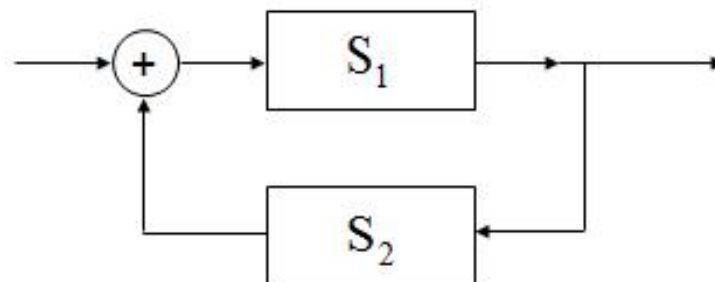
- **Series (cascade)**



- **Parallel**



- **Feedback**



System Properties

(Causality, Linearity, Time-invariance, etc.)

- Why bother with such general properties?
 - ◆ Important practical/physical implications. We can make many important **predictions** of the system behaviours without having to do any mathematical derivations.
 - ◆ They allow us to develop powerful tools (*transformations*, more on this later) for analysis and design.

1) Memoryless or With Memory

Memoryless : output at a given time depends only on the input at the same time

eg.
$$y[n] = (ax[n] - x^2[n])^2$$

With Memory

eg.
$$y[n] = \sum_{k=-\infty}^n x[k]$$

summer or accumulator

Memoryless or With Memory (cont.)

- In physical system, memory is associated with the storage of energy, e.g., capacitor in electric circuit.

2) Invertability

invertible : distinct inputs lead to distinct outputs, i.e.
an inverse system exists



No information loss

eg.
$$y[n] = \sum_{k=-\infty}^n x[k]$$

$$z[n] = y[n] - y[n-1] = x[n]$$

3) Causality

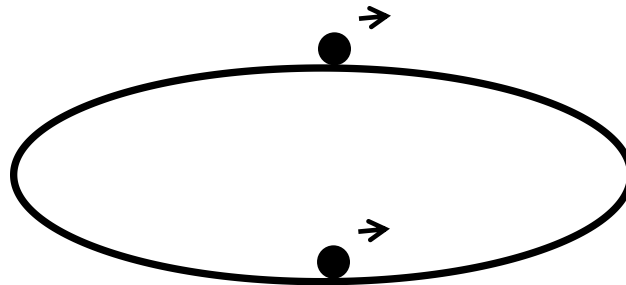
- **Causality**: A system is causal if the output does **not** anticipate future values of the input, i.e., if the output at any time depends only on values of the input up to that time.
- **All real-time** physical systems are **causal**, because time only moves forward, and effect occurs after cause.
(Imagine if you own a noncausal system whose output depends on tomorrow's stock price.)
 - ◆ Do **not** apply to spatially varying signals. (We can move both left and right, up and down.)
 - ◆ Do **not** apply to systems processing *recorded* (or *non-realtime*) signals, e.g. taped sports games vs. live broadcast.

Causal or Non-causal?

- $y(t) = x^2(t-1)$
- $y(t)=x(t+1)$
- $y(t)=x(t) \cos(t+1)$
- $y[n]=x[-n]$
- $y[n]=(1/2)^{n+1} x^3[n-1]$

4) Stability

- Small input leads to response that does not diverge.



- ◆ If the input to a stable system is bounded, the output must also be bounded.
- ◆ e.g.: $S_1: y(t) = t x(t)$
 $S_2: y(t) = e^{x(t)}$

Stability (cont.)

- Stability of physical systems are due to the presence of mechanisms that dissipate energy, e.g., resistor, friction.
- Stable or not (Example 1.13 in textbook)?
 - ◆ Find specific example, e.g, for S: $y(t)=t x(t)$
 - ◆ To prove that all bounded inputs lead to bounded output.

5) Time Invariance (TI)

- Informally, a system is time-invariant (TI) if its behavior does not depend on the choice of $t = 0$. Then two identical experiments will yield the same results, regardless the starting time.
 - Mathematically (in DT): A system $x[n] \rightarrow y[n]$ is TI if for *any* input $x[n]$ and *any* time shift n_0

$$\begin{array}{ll} \text{If} & x[n] \rightarrow y[n] \\ \text{then} & x[n - n_0] \rightarrow y[n - n_0] . \end{array}$$

- Similarly for CT time-invariant system

$$\begin{array}{ll} \text{If} & x(t) \rightarrow y(t) \\ \text{then} & x(t - t_0) \rightarrow y(t - t_0) . \end{array}$$

Time-invariant or Time-varying?

- Steps:

- 1) Calculate $y_1(t) \leftarrow x_1(t)$
- 2) Calculate $y_2(t) \leftarrow x_2(t) = x_1(t-t_0)$
- 3) Does $y_1(t-t_0)$ equal $y_2(t)$?

e.g.: $y[n] = \left(\frac{1}{2}\right)^{n+1} x^3[n-1]$

$$\textcircled{1} y_1[n] = \left(\frac{1}{2}\right)^{n+1} x_1^3[n-1]$$

$$\textcircled{2} x_2[n] = x_1[n-n_0]$$

$$\begin{aligned} y_2[n] &= \left(\frac{1}{2}\right)^{n+1} x_2^3[n-1] \\ &= \left(\frac{1}{2}\right)^{n+1} x_1^3[n-n_0-1] \end{aligned}$$

$$\textcircled{3} y_1[n-n_0] = \left(\frac{1}{2}\right)^{n-n_0+1} x_1^3[n-n_0-1]$$

$$\therefore y_1[n-n_0] \neq y_2[n]$$

$\therefore$ Time-varying

Now we can deduce something:

- If the input to a TI system is periodic, then the output is also periodic with the same period (Problem 1.43 (a)).

Proof: Suppose $x(t + T) = x(t)$
 and $x(t) \rightarrow y(t)$

Then by TI

$$x(t + T) \rightarrow y(t + T)$$



But these are
the same input!

So these must be
the same output,
i.e., $y(t) = y(t + T)$

6) Linear and Nonlinear Systems

- Many systems are **nonlinear**.
 - ◆ e.g.: economic system with
input: fiscal and monetary policies, labor, resources, etc.
→ output: GDP, inflation, etc.
 - ◆ System behaviour is very unpredictable, because it is highly nonlinear.
- We will deal with only **linear** systems, which are good approximations of nonlinear systems in certain ranges.
 - ◆ e.g.: small-signal conductance of a nonlinear diode.
- Linear systems can be analyzed accurately.

Linearity

We have $x_1(t) \rightarrow y_1(t)$, and $x_2(t) \rightarrow y_2(t)$.

The system is linear, if:

- 1) Additivity property: $x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t)$
- 2) Scaling (or homogeneity) property:

$$a x_1(t) \rightarrow a y_1(t)$$

where a is a complex number

e.g.: $y(t) = 2 x(t)$

$$y(t) = x^2(t)$$

Linearity (cont.)

A (CT) system is linear if it has the superposition property:

If $x_1(t) \rightarrow y_1(t)$ and $x_2(t) \rightarrow y_2(t)$

then $ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$

- For linear systems, zero input $\rightarrow$ zero output

"Proof" $0 = 0 \cdot x[n] \rightarrow 0 \cdot y[n] = 0$

Question: Is the system $y = A + x$ linear?

- *Incrementally linear system!*

Linear system or not?

- **Steps**

- 1) Have $y_1(t)$ and $y_2(t)$ as output signals to $x_1(t)$ and $x_2(t)$
- 2) Have $y_3(t)$ as output signal to $x_3(t) = a x_1(t) + b x_2(t)$
- 3) Does $y_3(t)$ equal “ $a y_1(t) + b y_2(t)$ ”?

More examples on textbook

Read Example 1.17 ~ 1.20

Linearity (cont.)

- Superposition

If $x_k[n] \rightarrow y_k[n]$

Then

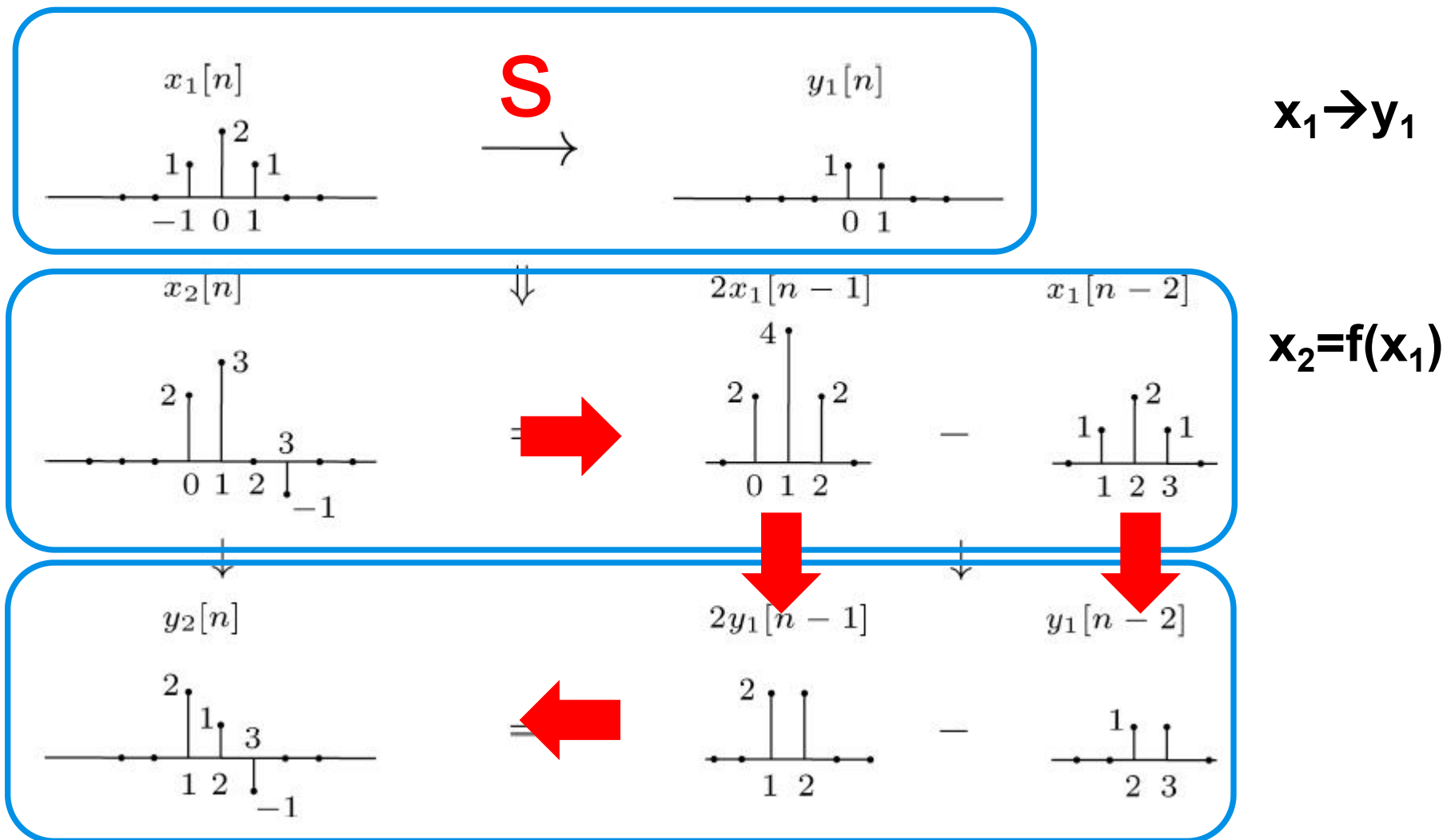
$$\sum_k a_k x_k[n] \rightarrow \sum_k a_k y_k[n]$$

- This property seems to be almost trivial now, but it is one of the most important ones

Linear Time-Invariant (LTI) Systems

- Focus of most of this course
 - ◆ Practical importance
 - ◆ The powerful analysis tools (*transformation*) associated with LTI systems
- A basic fact: If we know the response of an LTI system to **some** inputs, we actually know the response to **many** inputs.

Example: DT LTI System



Summary

- **Signals**

- ◆ Exponential signal (Difference between CT and DT)
- ◆ Unit impulse function
- ◆ Unit step function

- **Systems**

- ◆ Properties
- ◆ Linear, & time-invariant